

EAS 501 Lecture Notes 09/16/2026

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Outline

- Chapter 16. Scalar and Vector Field Theory

16.2 Preliminaries

scalar field: $f(x, y)$ (in 2-D), $f(x, y, z)$ (in 3-D),

vector field: $\mathbf{F}(x, y) = \begin{pmatrix} f_1(x, y) \\ f_2(x, y) \end{pmatrix}$ (in 2-D),

$\mathbf{F}(x, y, z) = \begin{pmatrix} f_1(x, y, z) \\ f_2(x, y, z) \\ f_3(x, y, z) \end{pmatrix}$ (in 3-D).

16.3 Divergence

For $\mathbf{F}(x, y, z) = (P, Q, R)^T$, we define

$$\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = P_x + Q_y + R_z.$$

Example 16.3 # 2. Compute $\nabla \cdot \mathbf{v}$

$$\begin{aligned} \mathbf{v}(x, y) &= U\mathbf{i} + \frac{Ua^2}{(x^2 + y^2)^2}((y^2 - x^2)\mathbf{i} - 2xy\mathbf{j}) \\ &= \begin{pmatrix} U + \frac{Ua^2}{(x^2 + y^2)^2}(y^2 - x^2) \\ \frac{-2Ua^2xy}{(x^2 + y^2)^2} \end{pmatrix}. \end{aligned} \quad (1)$$

Example 16.3 # 2, continued

We see that

$$\begin{aligned}
 \nabla \cdot \mathbf{v} &= P_x + Q_y \\
 &= \frac{Ua^2[-2x(x^2 + y^2)^2 - (y^2 - x^2) \cdot 4x(x^2 + y^2)]}{(x^2 + y^2)^4} \\
 &\quad - \frac{2Ua^2[x(x^2 + y^2)^2 - xy \cdot 4y(x^2 + y^2)]}{(x^2 + y^2)^4} \\
 &= 0.
 \end{aligned}$$

16.4 Gradient

$\nabla f = (f_x, f_y)^T$ in 2-D, and $\nabla f = (f_x, f_y, f_z)^T$ in 3-D

Given $f(x, y, z)$, $\mathbf{x}_0 = (x_0, y_0, z_0)$, $\mathbf{a} = (a, b, c)^T$, with $|\mathbf{a}| = 1$.

The **directional derivative** is defined as

$$D_{\mathbf{a}}f(x_0, y_0, z_0) = \lim_{t \rightarrow 0} \frac{f(x_0 + at, y_0 + bt, z_0 + ct) - f(x_0, y_0, z_0)}{t}.$$

Fact: $D_{\mathbf{a}}f(x_0, y_0, z_0) = \nabla f(x_0, y_0, z_0) \cdot \mathbf{a}$ for $|\mathbf{a}| = 1$.

Example 16.4 # 2 (c)

$u = f(x, y, z) = xyz$, $P = (1, -1, 2)$, $\mathbf{v} = (3, 0, -1)^T$.

Solution. $f_x = yz$, $f_y = xz$, $f_z = xy$, so that

$$\nabla f = (f_x, f_y, f_z)^T = (yz, xz, xy)^T,$$

$$\nabla f(1, -1, 2) = (-2, 2, -1)^T.$$

Example 16.4 #2 (c), continued

Meanwhile, $\mathbf{v} = (3, 0, -1)^T$ is not a unit vector, and we normalize it as

$$\mathbf{a} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{1}{\sqrt{10}} \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 3/\sqrt{10} \\ 0 \\ -1/\sqrt{10} \end{pmatrix}.$$

Finally, we get

$$\begin{aligned} D_{\mathbf{a}}f(1, -1, 2) &= \nabla f \cdot \mathbf{a} = \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 3/\sqrt{10} \\ 0 \\ -1/\sqrt{10} \end{pmatrix} \\ &= -2 \cdot \frac{3}{\sqrt{10}} + 2 \cdot 0 + (-1) \cdot \frac{-1}{\sqrt{10}} \\ &= \frac{-6}{\sqrt{10}} + 0 + \frac{1}{\sqrt{10}} = \frac{-5}{\sqrt{10}}. \end{aligned}$$

16.5 Curl

For $\mathbf{F} = (P, Q, R)^T$, define $\text{curl} \mathbf{F} = \nabla \times \mathbf{F} = \begin{pmatrix} R_y - Q_z \\ P_z - R_x \\ Q_x - P_y \end{pmatrix}$.

Fact: $\nabla \times (\nabla f) \equiv \mathbf{0}$

Example 16.5 # 1 (e)

$$\mathbf{v} = xy\mathbf{i} - 2(x^2 + z^2)\mathbf{k} = \begin{pmatrix} xy \\ 0 \\ -2(x^2 + z^2) \end{pmatrix}.$$

Solution. Since $P = xy$, $Q = 0$, $R = -2(x^2 + z^2)$, we have

$$\nabla \times \mathbf{v} = \begin{pmatrix} R_y - Q_z \\ P_z - R_x \\ Q_x - P_y \end{pmatrix} = \begin{pmatrix} 0 - 0 \\ 0 - (-4x) \\ 0 - x \end{pmatrix} = \begin{pmatrix} 0 \\ 4x \\ -x \end{pmatrix}.$$

16.6 Combination; Laplacian

- ▶ $\nabla \cdot (f\mathbf{v}) = \nabla f \cdot \mathbf{v} + f(\nabla \cdot \mathbf{v})$
- ▶ $\nabla \times (f\mathbf{v}) = \nabla f \times \mathbf{v} + f(\nabla \times \mathbf{v})$
- ▶ $\nabla \cdot (\mathbf{u} \times \mathbf{v}) = \mathbf{v} \cdot (\nabla \times \mathbf{u}) - \mathbf{u} \cdot (\nabla \times \mathbf{v})$
- ▶ $\nabla \times (\mathbf{u} \times \mathbf{v}) = \mathbf{u}(\nabla \cdot \mathbf{v}) - \mathbf{v}(\nabla \cdot \mathbf{u}) + \mathbf{v} \cdot \nabla \mathbf{u} - \mathbf{u} \cdot \nabla \mathbf{v}$
- ▶ $\nabla(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot \nabla \mathbf{v} + \mathbf{v} \cdot \nabla \mathbf{u} + \mathbf{u} \times (\nabla \times \mathbf{v}) + \mathbf{v} \times (\nabla \times \mathbf{u})$
- ▶ $\Delta f = \nabla \cdot (\nabla f) = f_{xx} + f_{yy} + f_{zz}$
- ▶ $\nabla \cdot (\nabla \times \mathbf{v}) = 0, \quad \nabla \times (\nabla f) \equiv \mathbf{0}$
- ▶ $\nabla \times (\nabla \times \mathbf{v}) = \nabla(\nabla \cdot \mathbf{v}) - \Delta \mathbf{v}$

Example 16.6 # 1 (e)

$$u = f(x, y) = xe^y$$

We have $u_{xx} = 0$, $u_{yy} = xe^y$, so that $\Delta u = u_{xx} + u_{yy} = xe^y$.

And also, we see that $\nabla u = \begin{pmatrix} e^y \\ xe^y \end{pmatrix}$, $\nabla \times \nabla u = \mathbf{0}$.